Fiscal Year:	FY 2007	**Task Last Updated:**	FY 06/19/2009
PI Name:	Belcher, James B.S.		
Project Title:	ElastiGlass Barrier Film and Food Processing Techniques for the 3-to-5 Year Shelf-Stable Food Package		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Space Human Factors Engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SHFH**:Space Human Factors & Habitability (archival in 2017)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	INDUSTRY	**Phone:**	(781) 890-1338
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PI Address 2:			
PI Web Page:			
City:	Waltham	**State:**	MA
Zip Code:	02451-1016	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	SBIR Phase II
Start Date:	01/12/2007	**End Date:**	07/01/2009
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Woolford, Barbara	**Contact Phone:**	218-483-3701
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 7/1/2009 per HRP Master Task List information dtd 1/11/2012 (Ed., 2/27/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNJ07JA14C		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The National Aeronautics and Space Administration (NASA) foresees extension of exploratory space missions to the Moon, Mars, Venus and beyond. To reach these outer locations will require development of a range of advanced technologies including life support systems. Food represents one of the most crucial components of life support, as the ability to supply safe, nutritious and organoleptically pleasing meals to space flight personnel for the duration of their mission will be of paramount importance in maintaining crew functionality and morale. In this Phase II Small Business Innovation Research (SBIR) program, Infoscitex Corporation will further develop an advanced barrier film, ElastiGlass, which will enable 3-to-5 year shelf life. This technology will also provide the added benefit of reducing the logistic burdens associated with waste handling, and minimizing package weight and storage space requirements. During this proposed program Infoscitex will modify the Phase I proof of feasibility barrier coating formulations to achieve the best combination of minimal oxygen and water vapor permeation, and maximized elongation properties to obtain the best of		

	product durability. Prototype barrier films and subsequent food storage pouches will be fabricated, filled with thermostabilizable foods and tested for shelf life capabilities. POTENTIAL NASA COMMERCIAL APPLICATIONS: Potential NASA applications include the projected utilization of proposed contract deliverables (e.g.,prototypes, test units, software) and resulting products and services by NASA organizations and contractors.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The impact that ElastiglassTM may have on the market is profound and our team will work to identify commercialization opportunities throughout the Phase 2 program. The NASA Food Technology Commercial Space Center (NASA FTCSC) is leading a national effort to develop foods and food-processing technologies that will enable space station and planetary space missions. Infoscitex and it's partners are well positioned to help NASA FTCSC achieve it's objectives through the continued development and commercialization of ElastiglassTM.
Task Progress:	New project for FY2007. Reporting not required for this SBIR Phase 2 project. (Note: task added to Task Book in June 2009.)
Bibliography Type:	Description: (Last Updated:)